

ARPIT YADAV

Full Stack Developer | AI & IoT Systems Engineer

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PROFESSIONAL SUMMARY

Results-driven Full Stack Developer and IoT Systems Engineer with proven experience architecting high-performance web applications, 3D interactive graphics, and embedded hardware safety prototypes. Creator of **AI Hub (arpit-fun.web.app)** and Smart India Hackathon (SIH) prototype contributor. Proficient in React 19, TypeScript, Python (FastAPI), Three.js, C++, and hardware telemetry. Adept at engineering scalable software with deterministic state logic, cryptographic integrity (NIST SHA-256), and automated CI/CD pipelines.

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript (ES6+), Python, C/C++, HTML5, CSS3, SQL
Frontend & 3D Web:	React 19, Vite, Next.js, Three.js (WebGL, GLSL Shaders), Tailwind CSS v4, Framer Motion, Web Audio API
Backend & Systems:	FastAPI, Node.js, Express, RESTful APIs, SQLite (FTS5), WebSockets, JSON Schema, Microservices
IoT & Embedded:	Arduino (ATmega328P), ESP32, SIM800L GSM Telemetry, I2C/SPI Protocols, MQ Gas & Flame Sensors, FSM Architecture
DevOps & Cloud Tools:	Git, GitHub Actions (CI/CD), Firebase Cloud, Vercel, Postman, Linux/Unix, Chrome DevTools
Core Competencies:	Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Cryptographic Hashing (SHA-256), System Design

TECHNICAL PROJECTS

AI HUB (arpit-fun.web.app) - 3D Discovery Station & Media Suite [Live Demo] [GitHub] React 19, TypeScript, Three.js, Tailwind CSS
High-performance 3D AI Discovery Engine & Zero-Redirect Multimedia Suite

- 3D Helical Engine:** Architected a 3D periodic helical navigation system using Three.js and custom GLSL shaders with depth-of-field disc blur and 2x Retina supersampling, sustaining a smooth 60 FPS.
- AI Synthesis Studio:** Integrated multi-provider LLM processing (Google Gemini, Groq Llama-3) with local offline heuristic NLP fallback, delivering real-time compression ratios and one-click Markdown exports.
- Multimedia Suites:** Engineered an in-app eBook reader (70K+ Project Gutenberg titles with EPUB downloads), 4K cinema player, and 24/7 ambient streaming with Web Audio visualizers.
- Cloud CI/CD:** Built automated dual GitHub Actions CI/CD pipelines deploying to Google Firebase and GitHub Pages, achieving 98+ Lighthouse scores across performance and accessibility.

CHRONOFACT - Digital Forensic Investigation Workbench [Live Demo] [GitHub] Python, FastAPI, SQLite FTS5, SHA-256, AST
Verifiable Evidence Reasoning Workbench with BSA 2023 Section 63(4) Compliance

- Legal Compliance & Chain of Custody:** Developed an investigation triage platform complying with Bharatiya Sakshya Adhiniyam (BSA) s.63(4), implementing NIST SHA-256 hash-chained tamper-evident audit logs.
- Temporal Hyper-Timeline:** Constructed an interval-based timeline engine ($[t_{min}, t_{max}]$) and deterministic inconsistency radar to detect alibi contradictions, time-zone anomalies, and metadata tamper traces.
- Zero-Hallucination Gate:** Engineered a mechanical AST byte-span citation verifier intercepting responses to ensure 100% verbatim proof against original seized evidence, eliminating AI hallucinations.

Smart Kitchen Safety & Automation (Embedded IoT Prototype) [Live Demo] [GitHub] Embedded C++, Arduino Uno, Python (FastAPI), GSM

Deterministic Industrial Hazard Prevention Prototype with GSM Telemetry

- Hazard Mitigation & FSM:** Programmed a deterministic Finite State Machine (FSM) on ATmega328P mitigating gas leaks (>400 PPM) and flame radiation with automatic exhaust and sprinkler relay interlocks.
- Cellular Telemetry:** Implemented SIM800L GSM cellular telemetry using Hayes AT commands to broadcast emergency SMS alerts to designated responders upon hazardous state transitions.
- Noise & Power Isolation:** Engineered hardware isolation using a 3A buck converter and 1000μF decoupling capacitors to suppress relay back-EMF spikes, eliminating I2C LCD bus crashes. Built companion Python FastAPI serial telemetry workbench.

Smart Helmet - IoT Road Safety & Accident Mitigation System Arduino/ESP32, MQ-3 Sensor, C++, GSM/GPS
Smart India Hackathon (SIH) Prototype Showcase Project

- Compliance & Interlock:** Developed a two-wheeler ignition-interlock prototype enforcing mandatory helmet wear detection and zero blood-alcohol concentration (BAC) prior to enabling vehicle ignition.
- Crash Telemetry:** Integrated vibration and impact sensors triggering automated emergency SOS broadcasts with real-time GPS coordinates via GSM cellular network upon collision detection.

EDUCATION

Axis Institute of Technology and Management - Kanpur, Uttar Pradesh, India 2025 - 2029 (Expected)

Bachelor of Technology (B.Tech) in Computer Science & Engineering

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering.

HONORS & ACHIEVEMENTS

- Smart India Hackathon (SIH) Participant:** Shortlisted to engineer hardware-integrated IoT safety devices under competitive nationwide evaluation.
- Continuous Problem Solving & DSA:** Active algorithmic problem solver across LeetCode & GeeksforGeeks, focusing on core data structures, graph traversals, and dynamic programming.
- Production Deployment:** Architected, built, and deployed live web platforms (arpit-fun.web.app, arpity.me) serving real users with custom DNS routing and automated CI/CD.